

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester VI Examination, 2013-14

Paper: GLB-601 — Palaeontology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem VI/227

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Discuss the salient features of Darwin's Theory of Organic Evolution with evidence in support. [14]

Question 2. Write short notes on **any four** of the following: [4 × 3½ = 14]

- (a) Index fossils
- (b) Development of an individual organism
- (c) Stem and cirri of crinoids
- (d) Shell coiling in gastropods
- (e) Chemical composition of brachiopod shells

Question 3. Describe **any two** of the following: [2 × 7 = 14]

- (a) Description and preservation of a fossil
- (b) Shell forms in nautiloids
- (c) Zooecia of a bryozoan

Question 4. Discuss the diagnostic morphological features of bivalves. [14]

SECTION — B

Question 5. Give a brief account of **any two** of the following: [2 × 7 = 14]

- (a) Siwalik vertebrates
- (b) Lower Gondwana flora
- (c) Derived and living fossils

Question 6. Describe the morphological characteristics of ammonoids. [14]

Question 7. Write short notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Stratigraphic significance of conodonts
- (b) Guard in belemnites
- (c) Umbilicus and columella in gastropods
- (d) Nature of the anterior margin of brachiopods
- (e) Types in taxonomy

SECTION — C

Question 8. Define the following in 2 to 4 sentences: $[7 \times 2 = 14]$

- (i) Prolegulum and prodissoconch
- (ii) Organic-walled microfossils
- (iii) Operculum
- (iv) Siliceous microfossils
- (v) Ammonitic suture with a neat sketch
- (vi) Calcareous microfossils
- (vii) Epirostrum (Epirostum)